Zhiyuan Ma

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EDUCATION

Shanghai Jiao Tong University, Shanghai, China *Sep 2020 – Jun 2026*

BSc in Physics, **GPA**: 3.81/4.30 (Class Ranking 6/23) **Honors:** Zhiyuan Honors Academic Scholarship (2024, 2025)

Relevant Coursework: Introduction to Computer Science (A); Introduction to Computational Physics (A+); Introduction to Computational Physics and Scientific Computing (A+); Linear Algebra (H) (A+); Mathematical Analysis (H) I (A); Mathematical Methods in Physics I (A); Biological Physics (A); Quantum Mechanics I (A+).

Online Coursework (self-study): Introduction to Neural Computation, MIT OpenCourseWare 9.40 (Spring 2018).

GRE 330 (Verbal 160, Quantitative 170, AW 3.5); **TOEFL 5.0** (Reading 5.5, Listening 6.0, Speaking 4.5, Writing 4.5)

RESEARCH

Circuit Modeling of Temporal Association Learning in Trace Fear Conditioning *Aug 2026 – Present*

Research Assistant, Supervisor: Prof. Douglas Zhou, Shanghai Jiao Tong University

- Synthesized behavioral, neural-recording, and circuit-perturbation findings into a validation framework linking stimulus responses, learning-dependent representations, and intervention effects.
- Developed a GPU-accelerated excitatory–inhibitory recurrent rate model of entorhinal cortex, CA1, and subiculum in Python, incorporating slow stimulus responses and eligibility-trace-based synaptic plasticity.
- Qualitatively reproduced cue- and trace-responsive activity patterns; decoded training history from simulated CA1 activity using linear SVMs and evaluated the effects of circuit perturbations during learning and retrieval.

Computational Analysis and Modeling of Spontaneous Activity in the Dunnart Cortex *Sep 2025 – Jun 2026*

Undergraduate Thesis | Shanghai Jiao Tong University & WashU Medicine
Supervisors: Prof. Douglas Zhou and Prof. Geoffrey Goodhill; Collaborator: Prof. Linda Richards

- Developed a Python pipeline for single-neuron GCaMP8m calcium imaging, combining CaImAn preprocessing, hexagonal lattice mapping, and 3D breadth-first search to detect splitting, merging, and propagating events.
- Identified coexisting patch-like and wave-like events within individual recordings; PCA indicated high-dimensional, distributed activity. These descriptive findings refined earlier region-specific accounts.
- Built a retinal-wave-inspired network model that qualitatively reproduced localized and broad bursts across coupling strengths. Virtual calcium imaging suggested that fluorescence decay, thresholding, and noise partly explain long activity tails and neighbor-activation asymmetry.

Modeling and Simulation of Large-scale Cortical Network Dynamics *Sep 2024 – Sep 2026*

Research Assistant, Supervisor: Prof. Douglas Zhou, Shanghai Jiao Tong University

- Developed a GPU-accelerated whole-brain dynamic mean-field model in Python with regional heterogeneity across 68 cortical areas; fitted 138 parameters using Levenberg–Marquardt optimization, achieving a validation functional connectivity correlation of 0.74.
- Identified approximate power-law scaling in Gauss–Newton eigenvalue spectra across three working points and validated the associated sensitivity hierarchy through systematic perturbations along all 138 eigenmodes.
- Demonstrated that the leading sensitivity mode was robust across independent seed panels at each working point, while higher modes were less reproducible; compared selected leading modes with cortical maps to explore biological relevance.

SKILLS

Programming & Tools: Python, MATLAB, Mathematica, CaImAn.

CONFERENCES & SCHOOLS ATTENDED

Society for Neuroscience Annual Meeting 2025 (San Diego, United States) *Nov 2025*

The 14th Computational Neuroscience Winter School (Shanghai Jiao Tong University, Shanghai, China) *Jan 2025*